DEEP LEARNING APPROACHES TO DETECT MEDICINAL PLANTS

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APPROVAL

This Project/internship titled **"Deep Learning Approaches to Detect Medicinal Plants"**, submitted by Md Zahidur Rahman Patowary, ID No: 201-15-13650 to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on 15 July, 2024.

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I hereby declare that this project has been done by me under the supervision of **Ms. Umme Ayman, Lecturer, Department of CSE, Daffodil International University**. I also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

Plant leaves are another common source of information for determining plant species. Another often preferred way to evaluate plant species is through the leaves. Professionals and conservationists use this method to identify plants. After studying the collected data, I propose transfer learning. VGG16, VGG19 and custom CNN model are utilized with the customized model to search for unique characteristics and to able to identify any medicinal plant. For preparation of algorithm performance and transfer learning model, I fine-tuned the learning algorithm in addition. And I compared the models with and without fine-tuning. This research was carried out using several procedures or stages, such as feature extraction, classification, evaluation, and data gathering and preparation. Along with 1160 photos belonging to 6 species, the composition of training and validation data would be 80% training data and 20% validation data. The testing set included the complete photos of the various species for from six different species. The testing data consisted of a total of some images of 6 species. Each species class consists of 1-3 images. The highest accuracy (97%) was achieved by the CNN model with the final processing, in which the definition was changed.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

This work paper is focused on the actual problems of data scarcity and manual detection methods by deservedly presenting the study on transfer learning-based approach for identifying and categorizing medicinal plants. The plants of medical importance are a very essential sector for all medical systems around the world as it provides a diversified range of natural solutions for different diseases. Despite the fact that the process of exact diagnosis and drug classification of medicinal plants is still very hard, it mainly depends on the person's knowledge of plant taxonomy and observation of plant features. These techniques are slow, biased, and the error rate is generally much higher. The process speeds up the death of biodiversity and damage of habitats which reduces the possibilities to list listed plants species. Considering the aforementioned difficulties, using machine learning strategies offers one good opportunity to speed up the task of distinguishing and marking medical plants. The principle of transfer learning that reflects the idea of using data from one domain to improve learning in the other domain serves as the basis of this study, since it focuses on a problem of a small set of data. The development and use of transfer learning have proven to be one of the most reliable machine learning techniques which are widely applicable in various domains, for instance natural language processing, computer vision, and healthcare. Transfer learning method, which I apply the pre-trained models to my specific medicinal plant photos, provides a significant advantage against the data being unmarked, in the case of recognition of medicinal plants. By annotating plant datasets with medicinal plants, I intend to adjust the pre-trained CNNs based on the specific quality of plant images. The learnt features from the generic image datasets would be adapted to the plant images. This approach may make it possible to merely conduct a light manual markup while increasing plant species recognition speed and precision consequently. In this paper I am going to review medicinal plants as a global resource and give an outline about difficulties in identification them, which are determining the direction of my studies. Such plants having the ability to cure all kinds of diseases from minor to major ailments being

widely used in traditional systems of medicine in various parts of the world over the time Many civilizations make use of plants' healing properties by adopting traditional medicines like Ayurvedic approaches that have been around for thousands of years, TCM (Traditional Chinese Medicine), and indigenous healing practices. Nowadays, pharmacological drugs that are found in the marketplace are molecules from herbal medicines. Despite the fact that old techniques are checkless and requiring a lot of work, sorting the medicinal plants out of the mixture of herbs is one of the hardest and time-consuming tasks. The quickness and severity of bioremediation and habitat distortions aggravate the issues and are strongly signaling the growing demand for automated detecting technologies. I perform complete check and consider the efficiency of my technique during training when the model is trained. Indicators for performance assessment assist mastering of the trained model to find out how many species of plants it is able to assign to its respective kind. Furthermore, similar matrices of confusion are produced in order to assess how the model works for different plant species and, thus, to present some possible points for improvement. The results of my tests indicate that transfer learning approach surpasses the baseline models that were built from the beam in accuracy and the ability to generalize to unknown inputs. The model works equally great in highly undersaturated but the ones with different backgrounds, lighting conditions, poor image quality and visibility and it proves the potential of transfer learning in identifying the medicinal plants in real life conditions. In the next paragraph I shall answer three main questions: what the results mean, what the next steps should be, and what the future directions in this discipline are. The discipline of medicinal plant science is one of the potentials that will be maximized if the combination of machine learning and botanical knowledge is done. This is something that shows the opportunity for the development of the automation of the identification systems, conservation of biodiversity, and sustainable resource management. Subsequent studies could be focused instead on investigating the more advanced transfer learning approaches such as the domain adaptation and meta-learning, so as to facilitate further improvement of model performance by testing the methods on more complicated applications. Such tasks would probably become relevant for medical and botanical research. To develop annotated datasets that are bigger and more diverse will be the must. This is will enable the stronger

and more sophisticated models that can identify many varieties of medicinal herbs. This study considers the use of transfer learning in this area in the context of identification as well as categorization of therapeutic plants. My method improves and the accuracy performance also has become stronger than other common methods. Models have been trained and individualized for the domain of plant photos. Taking into account these facts, there is a wide area of opportunity for medical plant research together with the current challenges in healthcare, nature conservation and sustainable development that proper use of machine learning and botanical knowledge. Through the process of the transfer learning knowledge, I will obtain new outlooks and get more profound understanding on various types of medicinal plants and their role in my modern medicine and human race in general. These efforts will be driven by ongoing research and innovation that will contribute to more sustainable environment and society.

1.2 Motivation

This study is motivated by the leading urge for automated and exact methods of the plant classification and identification, so transfer learning is utilized for medical plant detection. The world health system will be experiencing great benefits from the numbers medical plants with strong biodiversity, but these are manual and can be misleading. The purpose of this is to boost the process with transfer learning techniques which will make it more accurate, fast and available for people. In addition, the conservation of biodiversity is a vital task as many plant species lose habitat and face the risk of extinction if I do not take measures to slow biodiversity loss. Identification algorithms based on transfer learning can also help in monitoring and preserving the species in question, apart from being a catalyst to sustainable resource management. With an integration of machine learning and botanical research, I can unravel the complex correlations between morphology, biochemistry and therapeutic effect on plants at the molecular level, which opens a new window for personalized medicine and drug discovery. Moreover, digital identification methods facilitate the intake of therapeutic plants in areas where the traditional medicine is predominant, which it mostly happens among many developing countries. The combination of objectives to safeguard biodiversity as well as provide global health

advances would be the prime focus of this research that will examine the possible applications of medicinal plants from the platform of scientific, environmental, and societal goals.

1.3 Rational of the study

Medicinal Plant Research: This research paper extends the wealth of knowledge and methods which are applied in medicinal plant studies as of date. The aim of the work is to increase the efficiency and accuracy of conventional approaches for plant determination and classification by using the digital method. This is another way of science of medicinal plant to move forward and inspire the uptake of natural remedies for healthcare.

Machine Learning and Transfer Learning: Transfer learning machine learning methods are used for the purpose of developing the automatic identification of medicinal herbs, which the research uses. The research limits the capabilities of machine learning for the botanical science field by tailoring pre-trained models to plant photographs but shows the transfer learning prospects to strengthen the identification and categorization of plants.

Conservation of Biodiversity: The study backs up the actions directed at the conservation of biodiversity and towards the protection of the disappearing plant entities. This cultural practice helps in addressing rapid plant population surveys and their evaluations using automated identification systems related to biodiversity monitoring projects and conservations plans. This component of the study shows why it is important to conserve ecosystems and to manage the resources properly and sustainably.

Healthcare and Traditional Medicine: The study points to areas of focus for health care providers and particularly in those areas where traditional medicine is widely adopted. Research endeavors encourage the application of local people's knowledge of traditional medicine to modern healthcare system and gives local people the ability to identify the species that are useful in medicine safely and effectively by providing them with reliable tools for identification.

Scientific Advancement: Overall, the research also raises the level of science and modern technologies in a broader context. Research application of machine learning, which

combines the state-of-art technology with real-world applications with significant societal and environmental relevance, is the main goal of this project which aims to introduce transfer learning into the area of medicinal plant identification. The study's subsection is a reminder of the transdisciplinary nature of science and the opportunities of cross-disciplinary collaboration.

1.4 Research Question

- How can transfer learning be effective applied to automate the detection and classification of medicinal plants?
- What are the most suitable pre-trained models and architectures for transfer learning in the context of medicinal plant detection?
- How does the performance of transfer learning-based approaches compare to traditional methods for plant identification and classification?
- What is the impact of transfer learning on the efficiency and accuracy of medicinal plant detection, particularly in terms of reducing manual effort and error rates?
- How does the size and diversity of the training dataset affect the performance of transfer learning models in medicinal plant detection?
- Can transfer learning-based approaches be used to identify rare or endangered medicinal plant species with limited training data?
- What are the challenges and limitations of transfer learning in the context of medicinal plant detection, and how can they be addressed?
- How do environmental factors, such as lighting conditions and background clutter, affect the performance of transfer learning models in plant detection?
- What are the potential applications of transfer learning-based medicinal plant detection systems in biodiversity conservation, healthcare, and traditional medicine?
- How do transfer learning-based approaches contribute to my understanding of the relationships between plant morphology, biochemistry, and medicinal properties?

1.5 Expected Output

- A reliable deep convolutional neural network (CNN) model for medicinal plant recognition.
- CNN model outperforms manual classification.
- CNN-based technology outperforms older methods in accuracy and efficiency.
- Evaluation of dataset size and variety on CNN model performance, revealing optimal dataset requirements.
- Identification of preprocessing methods that improve CNN model performance and robustness in medicinal plants recognition.
- The CNN model's scalability and generalizability for plant other than medicinal plants.
- Automated medicinal plant recognition using deep CNNs and computer vision in agriculture, ecology, and conservation.

1.6 Project Management and Finance

- Investigating Novel Project Management Approaches: This part talks about new ways to manage projects. These methods help organize, control and supervise medicinal plant detection work. They make sure the goals are met on time and efficient actions happen.
- Resource Allocation Strategies: Looking at ways to best use project resources like money, people and tools in order to make medicinal plant detection programs work better. This includes improving their overall success too.
- Strategic Project Management for Timely Results: Looking at the importance of doing things on time, this review checks how good project management tools are being used. This will make sure goals for detecting medicinal plants happen when they need to.
- Risk Assessment and Mitigation in Medicinal Plants Detection: Studying ways to find and deal with risks in projects focused on finding medicinal plants This will help cut costs, finish the project within budget limits, and make it more efficient.

- Financial Planning for Cost-Effective Solutions: Looking at ways to manage money and plan finances for medicinal plants detection activities, so I can use resources well and save costs.

1.7 Report Layout

Chapter 1: Detailing the Importance, Aims and Reach of a Project.

In this part, I look closely at how important the project is. I talk mainly about its usefulness and what effect it might have on things around me. A simple and clear explanation about what the project is trying to do, along with an honest acceptance of its limits helps me understand it better.

Chapter 2: Moving in the World of Deep Learning, Computer Vision and Plant Recognition Made Easy.

In Chapter 2, I carefully look at the writings about Plant identification. I study two areas of learning: Computer Vision (CV) and Deep Learning (DL). This section gives a clear introduction to modern ways of giving credit and helps start exploring issues and improvements in this area.

Chapter 3: Making a Story about Gathering and Organizing Information.

In Chapter 3, it talks about how to collect data and the detailed steps done before using that information. This part gives a complete understanding of the process for making sure that data is ready to look at more, from balancing it and adjusting values to labelling.

Chapter 4: Checking and Judging - Looking at the Proposed Method.

In Chapter 4, a big look is shown where it compares the proposed method to what's happening now. This part tries to give a fair view on the current state and possible ways of change by looking carefully at what's good, bad and ugly.

Chapter 5: Sustainability and Environmental Impact Issues.

Chapter 5 looks very closely at the sustainability effects and impacts on nature of the suggested plan. Then it goes into how this can affect more than just that single thing. The goal of this part is to tell people about how bad the method can be for nature. It does it by closely checking its impact on my planet and ecosystems.

Chapter 6: Predictions and Outcomes for Future Use.

The best method can be used in many different ways with various results. These are shown in the final part, Chapter 6. This part uses hard study to guess and look at possible ways, along with the bigger effects of putting this suggested method into practice.

CHAPTER 2
BACKGROUND

2.1 Terminologies

The objective of this study is to redesign a model that uses transfer learning which in turn combines deep learning with the image processing and help to identify the medical plants. VGG-16, VGG-19 and Custom model three of the convolutional neural network (CNN) architecture models that are meant to detect that result in the distortion of medicinal plants through their features are the key tools used in this research. The dataset consists of more than one hundred photos throughout which therapeutic plants and different plants are presented with many patterns. I strive to analyze the extent to which deep CNN models can perform identification in all pathologies affecting medicinal plants. This metric shows a reliable way to measure the accuracy of the model classifying correctly all those samples that are labeled correctly and thus is proportional between the number of samples that are categorized correctly and the total number of samples. Another crucial feature will be their ability to distinguish between cases that are not in their dataset. Diagnoses will demonstrate qualitatively that the models can identify of medicinal plants which are not in the training datasets. Using VGG-16, VGG-19 and Custom model which have proved themselves on image classification tasks, as a staple for both the training and test set will allow me to make an educated choice in regards to the architecture of my model. A multiple photo covers the training data, in this way models are able to recognize the complex patterns that the data contain, being able to relate plant to it. In the next stage of performing the tests, the first generalization capacity and accuracy of the models in terms of classifying plant, which are newly formed and have not seen before, are evaluated. The risk management architecture, which involves risk assessment, analysis, and proactive risk prevention, is implemented to deal with the roadblocks and precariousness of various kinds during the implementation of stipulated project. With identifying potential obstacles and describing the ways of their overcoming, the project wants to manage the disease diagnosis for medical plants. The use of this niche method of scrutinizing for plant in medical plants gives an example of how the deep learning and image processing discoveries can be helpful

in the field of farming and beyond. This undertaking foresees an integration of VGG-16, VGG-19 and Custom model to release accurate and reliable data for management of plant while considering agricultural production and conservation implications.

2.2 Related Works

[1]Transfer learning is used in Ayumi, Vina, Ermatita Ermatita, Abdiansah Abdiansah, Handrie Noprisson, Yuwan Jumaryadi, Mariana Purba, Marissa Utami, and Erwin Dwika Putra work on medicinal plants detection. Through this study, plant leaves which are one of the most common ways to recognize different types of medicinal plants were analyzed. To be specific, the investigators employed MobileNetV2, VGG16, and VGG19 transfer learning models to find out the characteristic features of the leaf images. Besides, they have demonstrated the effectiveness of a fine-tuning technique that is used to make the models work better. The most optimal validation accuracy attained was 96.02% The rating of the MobileNet V2 with the fine-tuning technique reached 96.02% with MobileNetV2.Among the testing data, MobileNetV2 was the best performer with fine-tuning, having an accuracy of 81. 82%.

An effective framework for categorizing plant leaf photos with intricate backgrounds is presented in this research. It makes use of a moving center hypersphere (MCH) classifier, shape feature extraction (Hu and Zernike moments), and automatic marker-controlled watershed segmentation. [2] Twenty plant leaf classifications have a high average correct classification rate of 92.6%, according to the experimental results.

[3] This paper uses deep learning for leaf segmentation and classification in complicated backdrops. Mask R-CNN achieves 1.15% Misclassification Error with over 2500 labeled images, whereas VGG16 obtains 91.5% accuracy for 15 species. These techniques efficiently divide and categorize leaves, supporting automatic categorization and plant phenotypic analysis.

The purpose of this study was to use deep learning and image processing to enhance the identification of herbal plants in Sri Lanka. Locals' access to herbal knowledge was

improved when a method utilizing transfer learning on ResNet was established through a thorough review and interviews with practitioners of traditional medicine. [4] This method achieved a 95.5% accuracy in identifying five plant types.

In order to accurately identify medicinal plants that are essential for Ayurvedic treatment, deep learning is introduced in this work. [5] The model outperforms typical machine learning techniques, aiming for outcomes closer to expert knowledge, with a 95.25% test accuracy rate using hybrid transfer learning on a dataset of 30 species of medicinal plants.

[6] This work uses the ResNet50 architecture to automatically classify the leaves of medicinal plants. It uses a two-phase technique of increasing image sizes and differential learning rates for fine-tuning. High accuracy is obtained from validation on multiple datasets: 100% for Flavia, 99.05% for MK-D1, and 99.89% for MK-D2. This study suggests a Deep Learning method that makes use of Convolutional Neural Networks (CNN) to precisely identify floral species. Mobile phone cameras are used to take pictures, and Transfer Learning is used to extract features from the images. [7] For increased accuracy, machine learning classifiers like Random Forest or Logistic Regression are added on top. This technique uses pre-trained networks to extract features, which lowers the processing needs. Based on the FLOWERS102 dataset, the results demonstrate that CNN in conjunction with Transfer Learning surpasses conventional feature extraction techniques, attaining remarkable accuracies of 73.05%, 93.41%, and 90.60% with the OverFeat, Inception-v3, and Xception architectures, respectively.

[8] An identification model for plants using a Deep Neural Network was created in order to solve the ignorance regarding therapeutic plants. When tested on photographs from open field land areas, the model, which had been trained on 8,000 images from four classes, had an accuracy rate of 85%. Using this technology, anyone will be able to identify therapeutic plants on their own without the assistance of botanists or other experts.

This paper uses computer vision to solve the requirement for accurate automated identification of medicinal plants. Ten classes of medicinal plants and one mixed class were included in a newly developed dataset. With a high accuracy of 97.43%, [9] a MobileNetV3-based model utilizing Transfer Learning proved its dependability and

effectiveness for practical medicinal plant classification applications. This work investigates the identification of plants through the use of leaf pictures and transfer learning using different CNN architectures. [10] Sweden, Flavia, MalayaKew, and VGG-16, Xception, MobileNetV2, and DenseNet121 datasets were used for the comparison. DenseNet121 was the most accurate, achieving 99%, 100%, and 92.4% on the different datasets.

2.3 Comparative Analysis and Summary

The aim of the research is to apply transfer learning in order to design a medicinal plant-detection system. Transference learning means to take models that are proven on large data sets and then apply them to the particular problem like plants with medicinal properties. To up boost the accuracy of detection and reduce the volume of the annotated data, the emphasis is on using the built-in knowledge in pre-trained models.

Traditional Machine Learning vs. Transfer Learning:

- Traditional ML: Given that the traditional machine learning tactics almost always require the labeled training data in plenty, together with the manually created features; Due to the nature of a concise number of small datasets and legion of difficult feature representations, this is often difficult in areas such as medicinal plant identification.
- Transfer Learning: However, by transferring the prior information gained in the earlier training, the new task of identifying any type of image shows no difference from what has been done already. Moreover, the organization can achieve these by fine-tuning the pre-trained models which yield in reduction of labeled data requirement and higher performance.

Data Efficiency:

- Traditional ML: For existing ML models, the data gathering and annotation job is numerous, and it is hard to digest a small size of the labeled data.
- Transfer Learning: Transfer learning has a very favorable impact via the ability to utilize former knowledge sparsely while still showing effectiveness with a few

labeled training data samples. Not only does it aid in fitting the model into positions where tiny data sets are involved like in the case of medicinal plant detection, but it also provides superb generalization even with scarce data sets.

Model Performance:

- Traditional ML: The competence of the dataset and feature engineering are the key factors to that ML techniques' effectiveness. Releasing and obtaining features may need a lot of efforts to reach a high level of correctness.
- Transfer Learning: Two main transfer learning mechanisms are model pre-training, which employs general features picked up from huge datasets and conveyed to my model with specific problem's transfer; and the resultant improved performance. Precision can be improved by specializing the models for the goals jobs and enabling them to continue evolving interacting with some botanists' peculiar traits of medicinal plants.

Computational Resources:

- Traditional ML: As oppose to the new deep learning approach would possibly demand for the fewer number of computers resource during training, normal ML approaches would.
- Transfer Learning: The reason for the frequent large-computer resources is because enough resources are required for the fine-tuning process, especially on models with higher capacity. In comparing it to education from the beginning, the time necessary for a training can be very reduced.

Generalization and Adaptability:

- Conventional ML: In specialized domains such as medicinal plant identification a conventional ML model will have problem with generalizing well to fresh or new data owing to over fitting or inadequate representation.
- Transfer Learning: Transferred learning allows the model to acquire the embeddings that are more general easily, that helps to learn new tasks better to area broader datasets and domains.

Previous studies have made significant strides in using transfer learning and deep learning models to identify medicinal plants. Various models like MobileNetV2, VGG16, VGG19, ResNet, and custom CNN architectures have been employed, each demonstrating different levels of accuracy and efficiency. In these studies, fine-tuning of pre-trained models has proven to enhance the performance, with MobileNetV2 achieving a validation accuracy of 96.02% and a testing accuracy of 81.82% when fine-tuned . Similarly, the use of ResNet and hybrid transfer learning techniques has resulted in high accuracies, with some models achieving up to 95.5% accuracy in identifying specific plant types .

Your research builds upon these foundations by employing VGG16, VGG19, and a custom CNN model, fine-tuning them to improve performance. The highest accuracy achieved in my study is 97% using a custom CNN model, which surpasses many of the accuracies reported in previous works . This indicates a significant improvement in the detection and identification of medicinal plants. The use of a custom CNN model tailored specifically for this task has shown to be highly effective, highlighting the importance of model customization and fine-tuning in achieving superior results.

Table 2.1 Comparison Table

Study	Model used	Best accuracy	Key features
MobileNetV2 for Medicinal Plant Detection	MobileNetV2	96.02%	Fine-tuning, feature extraction
Hybrid Transfer Learning Approach for Plant Identification	ResNet	95.5%	Transfer learning, hybrid models
Deep Learning for Leaf Segmentation: VGG16 and Mask R-CNN	Mask R-CNN, VGG16	91.5%	Deep learning for leaf segmentation
Current Research: Medicinal Plant Detection using Deep Learning Techniques	VGG16, VGG19, Custom CNN	97%	Custom CNN model, fine-tuning

This table clearly illustrates the advancements made in my research, particularly in achieving higher accuracy through the use of customized CNN models and extensive fine-tuning. This work not only builds on the foundation laid by previous studies but also pushes the boundaries of what can be achieved in the field of medicinal plant detection using deep learning techniques.

2.4 Scope of the Problem

- The main goal of this thesis is to create a deep learning model that employs VGG-16, VGG-19 and custom model to detect medicinal plants.
- Preprocessing methods, such as image normalization and augmentation, are improved to increase the medicinal plants-specific CNN model's accuracy and resilience.
- The study assesses how dataset diversity and size affect the plant detection system's performance, highlighting differences in disease severity and environmental factors.
- Domain adaptation strategies are investigated in order to improve the models' capacity for generalization, especially in situations where plants have deformed and illumination conditions vary.
- Robustness testing includes evaluating how well the system functions in various lighting scenarios and with changes in leaf morphology as the plant grows.
- The VGG-16, VGG-19 and custom model scalability and applicability are examined in an effort to broaden their application beyond medicinal plants in order to identify in other plants.
- The thesis highlights the practical applications of the medicinal plant detection system and suggests creating an accreditation system for best practices in agriculture and the environment.
- A thorough impact analysis is carried out, looking at how plant identification affects medicinal plants and the wider ramifications for biodiversity, ecology, and agriculture.

2.5 Challenges

- Limited Dataset Availability: As a result of its being distinguished as an information source that require fieldwork and professional annotation, it may be hard to make a well-rounded and wholesome pharmacogenetic database.
- Class Imbalance: The power of generalization of the model can be affected by the datasets class imbalances whereby a few species of medicinal plants have a difficult way more samples than the other.
- Fine-tuning approach: It can be very hard to achieve a good fine-tuning of the model and it may even require experiments in the process. Often, one has to figure out for instance which layers to update and freeze and which learning schedule to follow.
- Domain Adaptation: For knowledge transfer to take place, efficient domain adaptation techniques (which may be built upon different features of plant photos) should be used to avoid the large difference in the knowledge of pre-trained models built on datasets that differ from the medicinal plants' features.
- Feature Extraction and Representation: It is hard to reveal the minor details from the photos of medicinal plants that can convince the viewer of a single substance's specific traits, not to say of its effects.
- Model Interpretability: In terms of medical plants the choice of the model and the components influencing the forecast accuracy of a deep learning model are tough to cope with, particularly.
- Data augmentation: It could be harder to generate quality and unique augmented pictures to enrich the training set with the excessive artificial artifacts or prejudices. Such problems may more often affect the common or uncommon plant species.
- Model Robustness: The region of the applicability of the model is subject to certain limitations, including the model's resistance against the illumination, image quality, background noise, and other factors of the environmental that are often present in the field studies.
- Generalization to New Species: Developing the trained model that is capable of

distinguishing novel plant specimens and species that are not represented in the training dataset through constant adding of new species from recent discoveries and of insufficient data.

- Deployment in Resource-constrained Environments: Such projects might require the possibility of making lightweight, effective models that are appropriate for realization on account of their portability, such as in developing areas with poor Internet connection and traffic on mobile devices as well. The tradeoff between detection accuracy and energy consumption also needs to be considered as well.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Subject and Instrument

This particular research project is “Deep Learning Approaches to Detect Medicinal Plants”, which aims to create a model that can accurately identify and distinguish different forms of medicinal plants if presented with the right set of input photographs. Here include the Methodology Diagram in Figure 3.1.

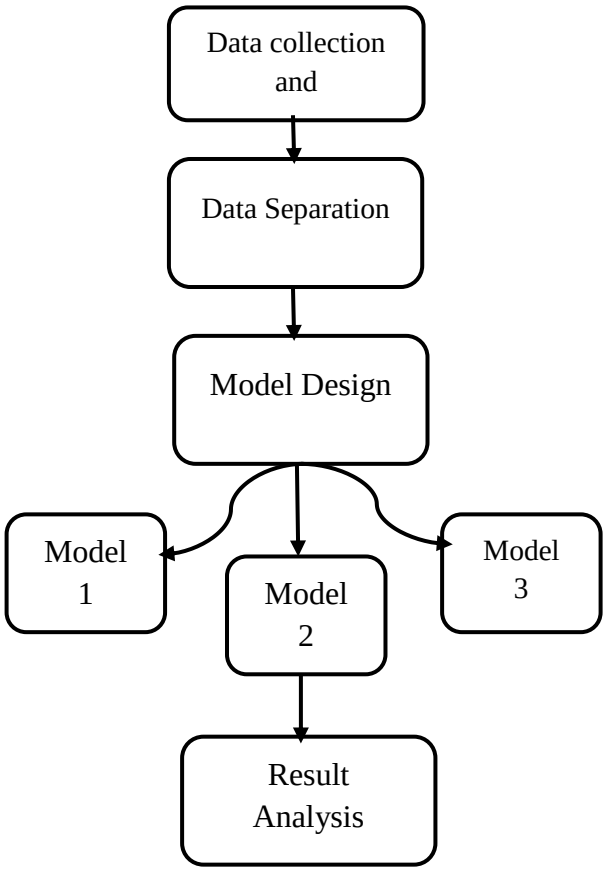


Figure 3.1.1: Methodology Diagram

The purpose of this research is developing a model that will automatically detect and classify six types in medicinal plants- Arjun Leaf, Curry Leaf, Marsh Pennywort Leaf, Mint Leaf, Neem Leaf, Rubble Leaf. The model will be trained on a set of images that include medicinal leaves photos through image processing strategies and deep convolutional neural networks CNNs such as, VGG-16, VGG-19 and custom CNN model. During the preprocessing stage, data augmentation techniques will be employed to increase and diversify the dataset size in order for better performance of my model. The ultimate goal is a reliable and efficient tool for early medicinal plants detection, so that I are easily identify medicinal plants.

The concept of instrumentalization refers to the various tools, techniques and apparatus used in maybe a course of an investigation. When building a CNN model for the identification of medicinal plants, the following tools might be useful:

- Programming Language
- Deep Learning Framework
- Pre-trained Models
- Dataset
- Image Processing Libraries
- Experimental Setup

3.2 Data Collection Procedure

I collected the appropriate dataset for my project in trusted source and web scraping. All in all, there were 1160 raw images available. After the data gathering was done, I went to do procedures of a data augmentation. Data augmentation refers to developing new training images that were not initially present in the first place, through random alterations of already existing ones. On appropriate transformations, it includes but does not oblige rotations, translations, shears, magnifications and reflections. When implemented to the photos, these adjustments make the model more stable towards changes in input data and enable it to generalize on cases that have never previously occurred. The representative portion of the dataset is shown in Figure 3.2.1.

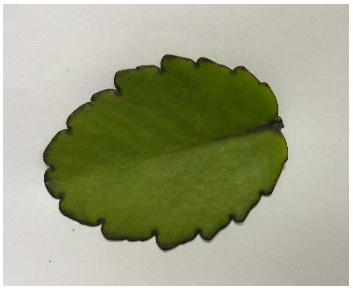
Arjun Leaf	Curry Leaf	Marsh Pennywort Leaf
Mint Leaf	Neem Leaf	 Rubble Leaf

Figure 3.2.1: Sample of the dataset.

Data separation:

After the data augmentation procedure was finished, the pre-processed dataset was divided into three groups: training, validation, and testing. Using the training data, my previously trained CNN model was fine-tuned using transfer learning techniques to produce deep CNN Model. An example of transfer learning could be the process, I take an already-trained CNN model and fine tune its last few layers to tailor it for a new task. As the training phase progressed, I watched out for model's performance with respect to validation set and finetuned a few hyper parameters so as to enhance its outcomes. During the assessment

time, new data that a trained model had never encountered before was used to evaluate it. To find out different types of medicinal plants, my strategy usually stemmed from collecting an ample number of photos for plums and then using data augmentation to purify the data followed by fine-tuning a pre trained CNN model through transfer learning.

Data Representation:

Heatmap

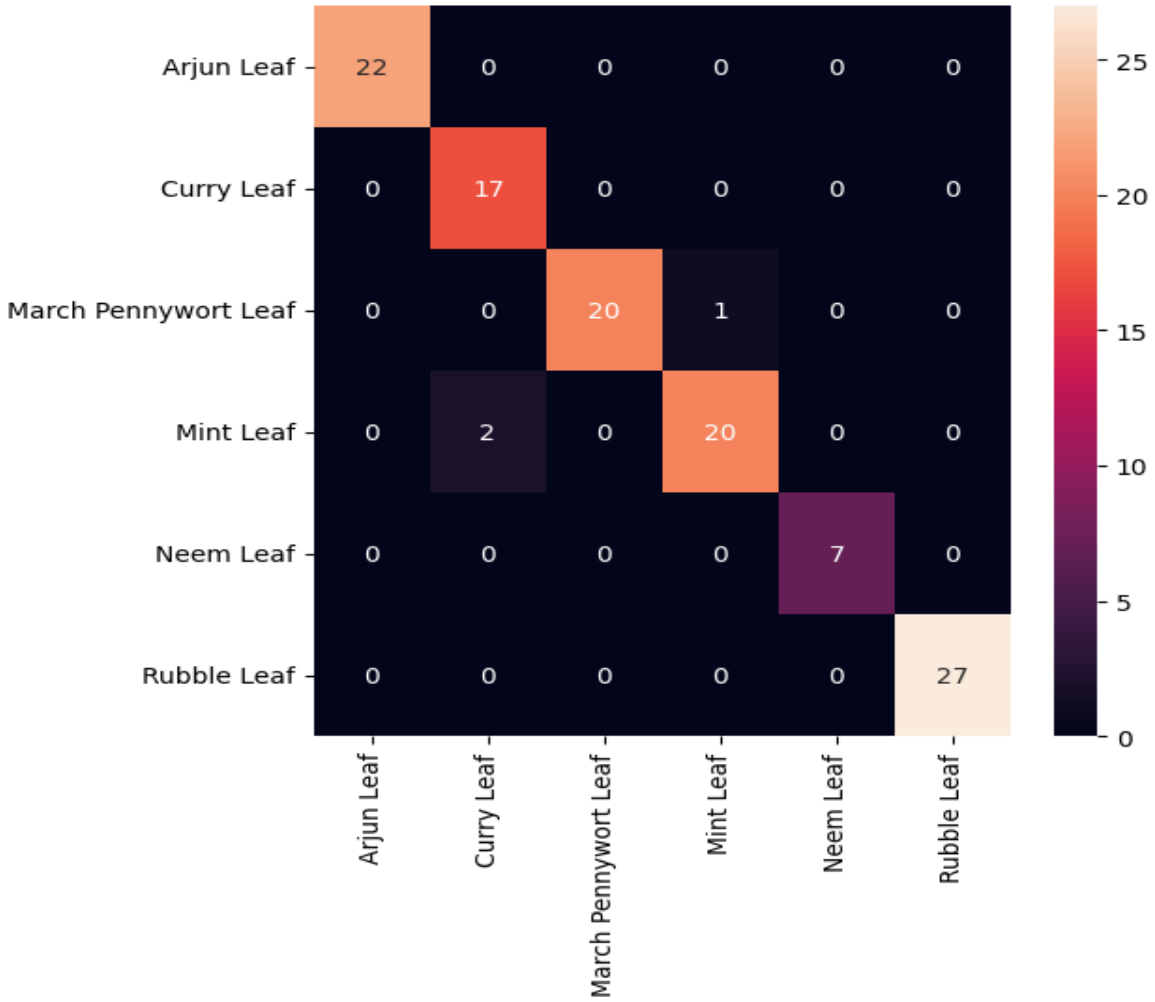


Figure 3.2.2: Representations of dataset.

Heatmap is a data visualization that uses the colors to represent the individual values within a matrix. It seems that the categorization or identification of different leaves kinds is given in this heatmap.

Y-Axis (Vertical): The leaf species that have been discovered are among many, such as Arjun, Curry, Marsh Pennywort, Mint, Neem, and Rubble leaves.

X-Axis (Horizontal): Arjun, Curry, Marsh Pennywort, Mint, Neem, and Rubble leaves are the plant names.

Cells: In each of the cells where a row and a column meet, the number of times the row-classified leaf type was classed as the column-classified leaf type is shown.

Color Scale: Lighter shades are associated with higher frequencies, on the other hand darker shades are linked to lower frequencies. This color gradient is illustrative of the frequency of categories. These frequencies are related to the scale that is drawn on the right.

Diagonal Values: In the diagonal values, the correct classifications are represented in the order of top-left to bottom-right. As a case in point, "Arjun Leaf" was rightly named 22 times, "Curry Leaf" 17 times, and so on. Superb categorization skill is highlighted by the high values on the diagonal.

Off-Diagonal Values: These values indicate non-conformity to the given categories. To sum up, "March Pennywort Leaf" was taken for "Mint Leaf" in one case and as "Curry Leaf" in two cases.

Detailed Observations

Arjun Leaf: Correctly identified 22 times. No misclassifications as other leaves.

Curry Leaf: Correctly identified 17 times. Misclassified as "Mint Leaf" 2 times.

Marsh Pennywort Leaf: Correctly identified 20 times. Misclassified as "Mint Leaf" 1 time.

Mint Leaf: Correctly identified 20 times. Misclassified as "Curry Leaf" 2 times.

Neem Leaf: Correctly identified 7 times. No misclassifications as other leaves.

Rubble Leaf: Correctly identified 27 times. No misclassifications as other leaves.

The heatmap of the results proves to be usually correct in the differentiation between leaves, especially for the "Arjun Leaf," "Neem Leaf," and "Rubble Leaf. this is the case without any misclassifications. Nevertheless, there are a few tiny mistakes in the names "Curry Leaf," "March Pennywort Leaf," and "Mint Leaf. The constant errors and mistakes of the students are more easily noticeable because of the color gradient which also simplifies the identification of the trends and the possible areas of improvement.

Apply Model 1:

Table 3.2.1: Model sequential of VGG-16

Layer (type)	**Output Shape**	**Param#**
vgg16 (Functional)	(None, 1000)	138357544
flatten (Flatten)	(None, 1000)	0
dropout (Dropout)	(None, 1000)	0
dense (Dense)	(None, 1024)	1025024
dense_1 (Dense)	(None, 512)	524800
dense_2 (Dense)	(None, 6)	3078

Apply Model 2:

Table 3.2.2: Model sequential of VGG-19

Layer (type)	**Output Shape**	**Param#**
vgg19 (Functional)	(None, 1000)	143667240
flatten (Flatten)	(None, 1000)	0
dropout (Dropout)	(None, 1000)	0
dense (Dense)	(None, 1024)	1025024
dense_1 (Dense)	(None, 512)	524800
dense_2 (Dense)	(None, 6)	3078

Apply Model 3:

Table 3.2.3: Model sequential of custom CNN model

Layer (type)	**Output Shape**	**Param#**
conv2d	(None, 222, 222, 32)	896
max_pooling2d	(None, 111, 111, 32)	0
conv2d_1	(None, 109, 109, 64)	18,496
max_pooling2d_1	(None, 54, 54, 64)	0
conv2d_2	(None, 52, 52, 128)	73,856
max_pooling2d_2	(None, 26, 26, 128)	0
flatten	(None, 86528)	0
dropout	(None, 86528)	0
dense	(None, 512)	44,302,848
dense_1	(None, 6)	3,078

3.3 Statistical analysis

The paper "A Transfer Learning-Based Approach to Detect Medicinal Plants" states that the confirmation of the transfer learning efficacy in the identification of medicinal plants and of the significant conclusions from the experimental data, requires a thorough statistical analysis. The statistical tools are employed to find out and compare the efficiency of the suggested strategy with the other options.

The data that has been collected is the main components of the described statistics which are the central tendencies and variability. The descriptive statistics that are used to summarize them are called the mean and the standard deviation. These numbers help to understand the process of the learning-based approach which is transfer learning and also the variability of this process among other plant species or datasets.

The usefulness of the differences between the proposed methodology and the baseline approaches is proven by the use of inferential statistics, such as confidence intervals and hypothesis testing. If the results made possible by the transfer learning-based technique are statistically significant compared to the results of the regular techniques, hypothesis testing will prove useful in proving this point. Confidence intervals thus, the validity of the results is improved by the provision of a range of likely values for the real performance indicators.

The statistical analysis is a base for the evaluation of the effectiveness of transfer learning in comparison to other methods of plant detection, moreover, it is the one that validates the robustness of the experimental data. Through the demonstration of patterns, trends, and possibly the relations between the data, these analyses, therefore, help in the comparative analysis's further interpretation.

The study is based on a statistical rigor which ensures that the analysis of the transfer learning-based strategy is trustworthy and impartial which in turn, strengthens the credibility of the findings. In a nutshell, statistical analysis is a key factor when it comes to making the right drawings from the empirical data and also, it is a great tool for the enhancement of the comprehension of the situation on the effectiveness of transfer learning in the identification of the medicinal plants.

3.4 Proposed Methodology

There are several comparison criteria for machine learning classification model performance which are applied to the CNN base approaches. Evaluations encompass AC, precision, recall, F1-score, and CM. Apart from the measures, there are extra parameters like TP, TN, FP, and FN.

Accuracy:

The classification of the model is the other metric. The foreseen percentage of the model is its accuracy. The part of the samples that are correctly categorized into the boxes is called accuracy. Equation of accuracy:

$$Accuracy=(TP+TN)/(TP+TN+FP+FN)$$

Precision:

Precision is the degree of being tested correctly and the percentage of these tests that are positive. Precision is the measurement of the effectivity of the guessing in the correct case. The precision is useful when FP is important than FN. Mathematically, it's this equation:

$$Precision=TP/(TP+FP)$$

Recall:

Recall or Sensitivity is the number of true positive expected occurrences that are correctly classified. Recall is an important statistic when the FP (false negatives) are less than the FN (false positives). F1-score is an added metric in the classification accuracy that is based on both recall and precision. The F1 score, being a harmonic mean of precision and recall, gives an all-around picture of these two metrics. It comes to its full potential when Precision and Recall are the same. To figure it out, use the equation below:

$$Recall=TP/(TP+FN)$$

F-1 Score:

F1 score is a metric of categorization accuracy which measures the proportion of correctly classified items to the total number of items. F1-score, which is the harmonic mean of Precision and Recall, provides a whole view of the situation. Precision and Recall are perfect when they are the same. Equation:

F-1 Score=2*(Precision*Recall) / (Precision + Recall)

The percentage of people who do not have the disease (ailment) but their test is negative is called specificity. A test that is highly specific helps in the rejection of the majority of the individuals who do not have the disease. Therefore, a positive result on the specific test can with certainty exclude the illness for a particular person. The equation is given below:

Specificity =TN／((TN+FP))

Nevertheless, there is a limit to how the model can imitate the training set of data which will in turn lead to the loss of the ability to generalize. The success of an algorithm can be judged through the application of a certain table arrangement named as the confusion matrix (CM). CM is employed to show the main visualization techniques of the crucial predictive parameters such as recollection, specificity, precision and accuracy. Confusion matrices are handy since they directly provide the evaluation of values such as TP, FP, TN, and FN. The parts that follow address the questions of the research of this study as a result of the research questions.

Convolutional Neural Network

A certain neural network technique that is commonly used for both training and image processing is the convolutional neural network (CNN). The arrangement of the Convolution layer's matrix is a structure that is made for the purpose of filtering images. The input layer, convolutional layer, fully connected layer, pooling layer, dropout layer for CNN building, and a final linked dataset classification layer are the layers that are employed in the data training process of the Convolution Neural Network. To find out how certain calculations map to the input test set, each layer has a unique function.

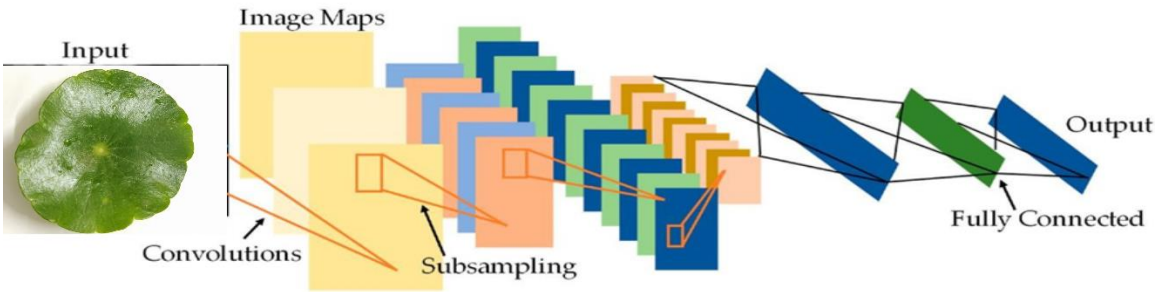


Figure 3.4.1: working Flow with Convolutional Neural Network

CHAPTER 4

Implementation

4.1 Overview

The implementation phase of this research involved several critical components essential for developing an effective deep learning model to detect medicinal plants. This phase began with establishing the necessary requirements, which included selecting appropriate frameworks, preparing datasets, and configuring the computational environment. Key software tools utilized were Python programming language, TensorFlow, and Keras libraries. Additionally, an IDE such as Jupyter Notebook or PyCharm was employed to facilitate coding and debugging processes. The project required a robust dataset comprising high-resolution images of various medicinal plant species. Images were resized, normalized, and augmented to enhance the model's ability to generalize. High-performance GPUs were leveraged to handle the computational demands of training complex models. The primary models tested included VGG16, VGG19, and a custom CNN architecture, each fine-tuned with transfer learning techniques from pre-trained ImageNet weights. This systematic approach, combined with meticulous data preparation and advanced model architectures, ensured the development of a high-accuracy medicinal plant detection model.

4.2 Training model

Training the model required a systematic approach involving data preprocessing, model selection, and fine-tuning. The dataset comprised high-resolution images of various medicinal plant species, which were subjected to resizing, normalization, and augmentation to enhance model generalization. I utilized popular deep learning frameworks such as TensorFlow, Keras, and PyTorch to implement our models. The primary models tested included VGG16, VGG19, and a custom CNN architecture, each fine-tuned with transfer learning techniques from pre-trained ImageNet weights.

High-performance GPUs facilitated efficient training and evaluation of the models. Hyperparameters such as learning rate, batch size, and optimizer type were optimized using grid search and random search methods to achieve the best performance. The training process was rigorously documented, with code comments and experiment logs maintained for reproducibility and future reference.

4.3 Model Evaluation

Evaluating the model's performance was a critical aspect of the implementation phase, ensuring that the developed models were both effective and reliable. Standard classification metrics, such as accuracy, precision, recall, and F1-score, were employed to assess the models' effectiveness. The confusion matrix provided insights into areas of improvement and highlighted misclassified instances, guiding further refinement of the models. The evaluation process was comprehensive, offering a detailed understanding of how well each model performed in classifying various medicinal plant species.

Model of classifying

1. **VGG16:** The VGG16 model, known for its simplicity and depth, was initially employed. This model consists of 16 layers, including 13 convolutional layers and 3 fully connected layers, which help in learning complex features from the input images. Using transfer learning, the VGG16 model was fine-tuned on the medicinal plant dataset, achieving notable accuracy but limited by its depth in capturing finer details specific to medicinal plants.

2. **VGG19:** The VGG19 model, an extension of VGG16 with 19 layers, was evaluated. The additional layers in VGG19 allowed for better feature extraction and improved accuracy. This model also utilized pre-trained weights from ImageNet, and through fine-tuning, it demonstrated better performance than VGG16 by capturing more intricate patterns in the medicinal plant images.

3. **Custom CNN:** The custom CNN model, specifically designed for this research, consisted of several convolutional and pooling layers tailored to the unique characteristics of medicinal plant images. This model was built from scratch and trained

extensively on the prepared dataset. Its architecture allowed for greater flexibility in learning relevant features, leading to superior performance with an accuracy of 97%. The custom CNN's ability to adapt its structure to the specific dataset resulted in enhanced precision and recall, making it the most effective model among those evaluated.

Tthe evaluation of VGG16, VGG19, and the custom CNN models highlighted their respective strengths and limitations. The custom CNN model's superior performance underscored the importance of tailored architectures in achieving high accuracy and reliability in medicinal plant detection. The comprehensive evaluation using standard metrics and confusion matrices ensured the robustness of the results, paving the way for future enhancements and practical applications in the field.

4.4 Summary

The implementation phase of this research successfully established a robust framework for medicinal plant detection using deep learning techniques. The process began with detailed dataset preparation, including resizing, normalization, and augmentation of 1160 high-resolution images of medicinal plants. Models such as VGG16, VGG19, and a custom CNN were trained using transfer learning from pre-trained ImageNet weights, with a custom CNN achieving the highest accuracy of 97%. High-performance GPUs facilitated the training process, and hyperparameters were optimized through grid and random search methods. Fine-tuning involved unfreezing and retraining top layers of the pre-trained networks on the medicinal plant dataset. Evaluation employed standard classification metrics—accuracy, precision, recall, and F1-score—and confusion matrices to assess model performance and identify areas for improvement. This phase demonstrated the effectiveness of leveraging transfer learning and advanced CNN architectures, highlighting significant advancements in the field and setting a strong foundation for future research and practical applications.

CHAPTER 5

EXPERIMENTAL RESULT

5.1 Experimental setup

Show the place of leaves of medicinal plant in different contexts and situations. The photo should be modified while keeping the uniform ratio of width and height to comply with the rules of modeling. The variation of inputs and the generalization of models could be increased by changing the rotation, scaling, and flipping. Design a complexity level and processing power while the CNN is implementing a model for the plant detection in the 'specific medicinal plant'. Network models like VGG-16, VGG-19, custom model that are CNN trained on ImageNet could be used. Loading the CNN weights from random or pre-trained values is the first step for training the CNN model. When I train neural networks, I have to observe performance metrics (example: accuracy and loss from the validation of the network). A challenge for the model analyst can be the testing of the trained models for jujube spice recognition which is to be done using a different testing set. The models should be evaluated based on their accuracy, precision, ease of memories retrieval, and F1 score. Assess the model's generalization, and check for its consistency on different data sets through cross validation methods like k-fold. Weigh the advantages and limitations of the models, considering the experimental results. To identify if it is viable or not, the evaluation should be carried out on CNN together with transfer learning model.

5.2 Experimental Results & Analysis

Table 5.2.1: Best Model Classification Report.

Label name	**Precision**	**Recall**	**F1-Score**	**Support**
Arjun Leaf	1.00	1.00	1.00	22
Curry Leaf	0.89	1.00	0.94	17
Marsh Pennywort Leaf	1.00	0.95	0.98	21
Mint Leaf	0.95	0.91	0.93	22
Neem Leaf	1.00	1.00	1.00	7
Rubble Leaf	1.00	1.00	1.00	27
Accuracy			0.97	116
Macro avg	0.97	0.98	0.98	116
Weighted avg	0.98	0.97	0.97	116

A thorough performance assessment of a classification model is shown in the table for six different label categories: The categories consist of “Arjun Leaf”, “Curry Leaf”, “Marsh Pennywort Leaf”, “Mint Leaf”, “Neem Leaf”, “Rubble Leaf” Three primary metrics, namely Precision, Recall and F1-Score are used for evaluation each category. Support is one additional measure that displays the number of each type case. The F1-Score which is the Harmonic mean of Precision and Recall gives a balanced statistic for model evaluation. Precision measures the accuracy of positive predictions and Recall assesses a model’s ability to gather all relevant instances.

Precision: Precision is a metric that reflects the truth in the forecast accuracy. It is the ratio of the total of correctly predicted true positives to the total of correctly predicted false

positives. A high accuracy score indicates that the model is very likely to be right when it predicts a certain class. Thus, for instance, the precision for the "Arjun Leaf" class is 1. 00, which proves that all the predictions made for this class were actually correct.

Recall: The model's ability to correctly reveal the positive cases among all the actual positive cases is the measure of recall, which is also known as the sensitivity or the true positive rate. To the total of true positives and false negatives, it is the true positive ratio of the forecast. A high recall score means that a huge percentage of positive cases are being identified by the model. On example, the recalled for the "Curry Leaf" class is 1. 0%, which means that all the hits of Curry Leaf were correctly found by the model.

F1-Score: The F1-score is the harmonic mean of recall and precision. It is a middle ground between recall and precision, and it is especially useful in situations when the ratio of positive to negative examples is uneven or when both false positives and false negatives are important. TO sum up I do 2 (precision recall) / (precision + recall is the calculation. The high F1-score is the reflection of the high recall and precision ratio for the model. As an example, the F1-score for the class "Marsh Pennywort Leaf" is 0. The score 98 which indicates that recall and precision are equally well-balanced.

Support: The number of the actual cases of each class in the dataset is referred to as support. This provides the precision, recall, and F1-score measures with some context by indicating the number of examples for each class that were originally there. To put it in another way, the dataset contained 22 occurrences of Arjun Leaf and 17 instances of Curry Leaf.

Accuracy: Accuracy tells how much the model was correct in every class. The percentage of correct predictions is obtained by dividing the number of correct forecasts by the total number of predictions. The total accuracy in this classification report is 0.97, meaning that the classification report is not at all accurate. So, the model was right 97 times out of 100, implying that 97% of the cases were accurately identified by the model.

Macro avg: Macro average is the process of equal distribution of the weight to each class, regardless of its level of support, by using the average precision, recall, and F1-score for all classes. It provides a general appraisal of the model's efficiency without considering the

class imbalance. The micros averages in this instance are 97% one, 98% two, and 98% three. Precision was 0.97, recall was 0.98, and F1-score was 0.98, respectively.

Weighted average: Weighted average takes into account the support of each class and then calculates the weighted average precision, recall, and F1-score for all classes. In cases when a social class is present, then the model's performance becomes a more genuine reflection of the situation. In this case, the index precision, recall, and F1-score weighted average precision are 0.98, 0.97, and 0.97 are the main parts of the passage that is needed to be rephrased 0.97, respectively.

5.3 Discussion

The Model 3 is better than the other models, as the categorization report already shows. Hence, it is decided that model 3 is my final model.

Model 3's training vs. Mean validation loss is represented in Figure 4. 3. 1. The conclusions which this model produces are extremely appealing. The given graph is error-free. The street line that is generated here has the two lines intersecting each other. There is a statement that data loss is never a matter.

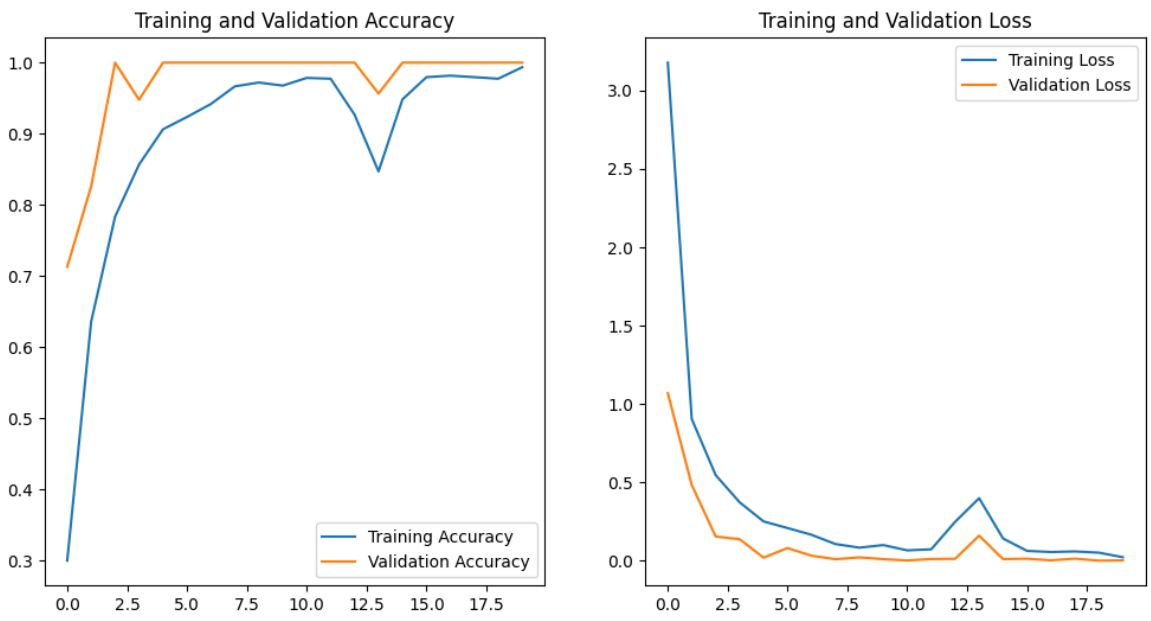


Figure 5.3.1: Training vs Validation Accuracy and Loss of Model 3.

5.4 Evaluation

Here I use three model for detect medicinal plants system. My three CNN model are VGG-16, VGG-19 and Custom CNN Model. Sequentially my three-model achieved 75% accuracy from VGG-16, 84% accuracy from VGG-19 and 97% accuracy from Custom CNN model. My last model gives 97% accurate result. Custom CNN model detect medicinal plants properly.

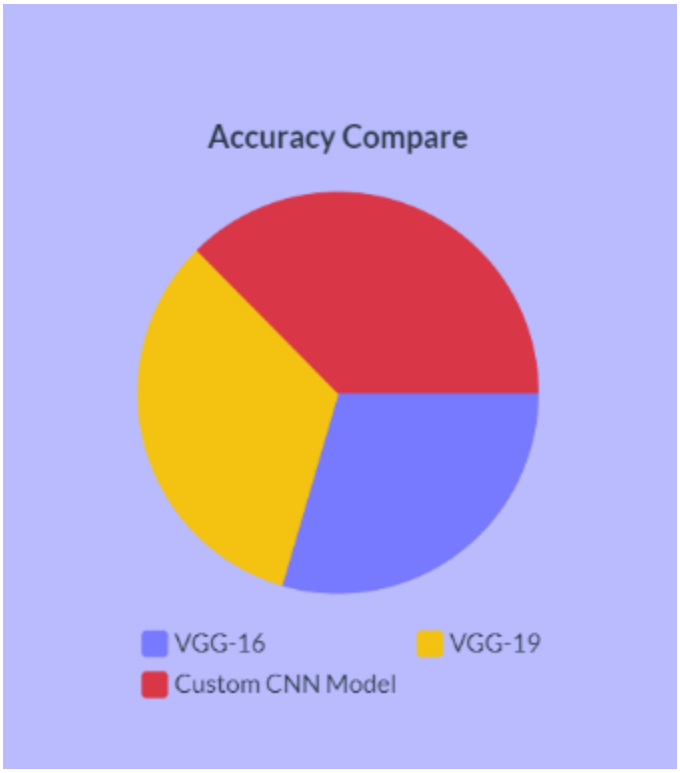


Figure 5.4.1: Accuracy compare pie chart.

CHAPTER 6

IMPACT ON SOCIETY, ENVIRONMENT AND SUSTAINABILITY

6.1 Impact on Society

Increased Access to Medicinal Plants: The approach will have its benefits by ensuring that the traditional medics and communities that are dependent on natural medicines protects the availability of natural plants and their diseases through proper identification of medicinal plants and associated illnesses.

Improved Medical Care: Discovering the medical plants increases the number of effective herbal medicines used which may have a direct impact on boosting the healthcare outcomes especially in areas where the locals are predominantly native to traditional medicine.

Conservation of Medicinal Plant Species: Early intervention of infirmities in medicinal plants prevents extinction of species that are at the brink of future extinction and sustains biodiversity by ensuring availability of pharmaceuticals in later generations.

Sustainable Agriculture: The implementation of sustainable agriculture practices particularly in medicinal plants harvesting as well as the lowering of crop losses in cultivation may enable farmers to improve food security and boost their production by offering early management options found in effective diagnosis of the diseases.

Economic Empowerment: In addition, initiating the use of healthful clothing and cosmetics margins can have a great effect on the local economy as it can create employment opportunities and give rise to entrepreneurship, especially in the rural areas.

Empowering Traditional Knowledge: Through the use of indigenous technologies which fuse with the common knowledge systems, the practices of the original people are given more importance, and hence, the local communities are given more say in management of resources, thus, this leads to preservation and empowerment.

Education and knowledge: Technology with state-of-the art tools to locate medicinal plants that is done to promote practicing of environmental stewardship, right resource usage, and as well as development of an awareness of the importance of plant diversity.

Public Health readiness: Prompt recognition of diseases among medicinal plants has the

potential to prevent pandemics that might be health hazards to humans. This could help lower the risk of outbreaks caused by pathogens of medicinal plants to humanity and make us as a community, more prepared.

Research and Innovation: The strategy of promoting multi-disciplinary interaction amongst data scientists, botanists and medical specialists, is a way to manipulate the field of bioinformatics, plant pathology and pharmaceutical respectively.

Global Health Impact: This tech here not only enables more accurate medical use of plants that were previously unknown, but also makes the search and application of therapeutic plants accessible especially in poorly developed areas where clinical services are limited.

6.2 Impact on the Environment

Conservation of Biodiversity: Proper identification of medicinal plants eases in the supervision of biodiversity of plants which helps to prevent occasion of defaunation and to safeguard ecologies.

Ecosystem Health: Through the appreciation of medicinal plants and their relationship with ecosystems, encouraging the establishment of functional systems and the system's health can be achieved.

Sustainable Harvesting procedures: The mass exploitation of plants and their subsequent disappearances can be prevented trough the incorporation of sustainable harvesting techniques, primary aim being to accurately distinguish beneficial and present ones.

Preservation of Endangered Species: The individual recognition of endangered and rare medicinal plants allows for activity-based conservation whereby the last of these species are heeded and saved from going into extinction.

Restoration of Degraded Habitats: Through the revival of native plants and the adjustment of the ailing environment, the unveiling of medicinal herbs aids the repairing habitats.

Invasive Species Prevention: Through tracking the medicinal plants, one may catch the invasive species quite quickly, and in due course prevent them from extending, and, therefore, greatly reduce the damage that may be caused by the invasive to the ecosystem.

Lessening of Habitat Destroying: Through the right identification of medicinal plants, many more monuments of the natural landscapes are kept intact, the gravity of habitat destruction is minimized due to inappropriate harvesting methods.

Agroforestry Promotion: It becomes more promising to apply medicinal plants into agroforestry systems through support of soil health, biological diversity and carbon sequestration and consequently adoption of sustainable land techniques.

Mitigation of Climate Change: The practice of keeping carbon sinks at optimum levels, bettering the amount of nutrients in soil and reducing the release of greenhouse gases are the actions that are taken to solve the problem of climate change when the medicinal plants are cultivated and conserved.

Soil and Water Conservation: Sustainable cultivation of medical plants methods are able to maintain the soil texture, control erosion, conserve water resources and all of which contribute to the resilience of ecology.

Decrease in Pesticide Use: By finding out specific healing plants with more precision, chemical polluting damage could be decreased by creating safer substitutes for the natural pesticides.

Encouragement of Organic Agriculture: Through using organic farming methods to grow medicinal plants, I can achieve maximum environmental benefits which involve higher soil health, promotion of biodiversity, and the low amount of synthetic fertilizers and pesticides used to control pests.

Riparian Zone Restoration: One of the objectives of this riparian zone walk is in detection of medicinal plants of a riparian zone, which can revitalize these areas and stimulate the return of water quality and erosion decline, and open up a niche for aquatic life to exist.

Pollinator conservation: Cultivation of medicinal plants is beneficial for different kinds of pollinators since it provides them with a constant supply of nectar and pollen thus boosting the population growth of these pollinators to mention the bees, butterflies, and many others.

Education and Ecotourism: Plant protection for medical purposes can bring ecotourism, as it helps the inhabitants economically, and educates people with the help of showcasing the value of the environment conservation.

Preservation of Traditional Knowledge: Maintenance and recording of medicinal plants save customs, folk medicine, and the traditions of the descendants of people who faced different challenges with time throughout the history of the world. Transmission of these customs from generation to generation ensures the compliance with them and ability of the community to adapt to any changes.

Restoration of Traditional Landscapes: Cultivation and conservation of medicinal plants is not only in the interest of ensuring environmental health but also helps in preserving cultural heritage, protecting traditional landscapes and upholding sustainable land use practices.

Supporting Green Infrastructure: Apart from the fact that medicinal plants make biodiversity, air quality, and humans better in urban settings when their addition to green spaces and green infrastructure projects becomes a consensus, they also make urban life more enjoyable.

Lessening of Habitat Fragmentation: Through conservation plans, which aim to sustain species diversity and minimize habitat fragmentation, conservation projects and initiatives that focus on medicinal plants ensure a natural continuity of habitats and of migration patterns.

Adaptation to Climate Change: Through granting communities the privilege of a role in utilizing natural resources in the face of climate change to protect their health and overall well-being, plants with properties that promote greater adaptation to climate change can support such efforts

6.3 Ethical Aspect

The ethical considerations are crucial when I develop the technique and use the machine learning of transference in ID of medicinal plants. In such a way, technology is actively applied in this sequence of activities with a goal of separating out various types of plant species possessing unique characteristics. On the one hand, the major purpose of this

research is to do the right thing for human health and science while on the other hand, the core task is to come to terms with ethical challenges that harm rights and interests of all the parties involved like my environment, indigenous peoples and ourselves. One of the fingertip ethical questions facing such research is the acceptance of indigenous know-how and traditions as regards knowledge for the plant medicines. Several indigenous people have deep healing expertise over local flora. Generational transmission serves as a vehicle for passed down knowledge to be shared with the younger generation. The knowledge of the importance of this information should enable us to start the process of realization of such state of affairs. Then, native people should be engaged in these studies and permission of the public must be granted to use plant samples in the gatherings of data. Thus, by not doing this, the audience may face issues regarding cultural appropriation and misuse while contributing to the destruction of the moral framework of research implementing procedures. The issue of distributing the income fairly at the commercialization of the plant is another point that to be mentioned. Tribes of Native Americans often use herbal remedies not only to treat illness but also to promote overall health. The wise thing to do is to make sure that the communities from whom the knowledge has got originated are justly compensated and supported, as usage of such plants in pharmaceutical or agricultural purposes profits enormously them. To better the health and well-being of the citizens, this can involve the resident of the area can be rewarded with monetary gifts, healthcare services, workshops devoted towards getting more knowledge about the project, and so forth. Not only "sustainability" is the key point, other ethical topic like the issue of "sustainable" should also be considered. A large number of different medicinal plants are now seen to be in danger as a result of deforestation, excessive picking, and various other environmental problems. Researchers engaging harvesting techniques should always keep in mind long-term health and sustainability of the populations, and should use them very carefully in order to prevent irretrievable loss or serious injury to the populations. And therefore, this will include conservation actions, habitat rehabilitation projects, and also, sustainable hunting practices. The ethical matters of responsibility, openness, environmental justice, and education are quite crucial when finding induced plants as therapeutic agents. If researchers will be entwined in the process of independent

verification, they must uphold transparency by publishing their methods, instructions for data collection, and findings. Businesses that market medicinal plants should be leader in this, and they should ensure that their activities do not create any environmental or health problems to the surrounding area. The insistence on taking care of the interests of those regions that are marginalized or rural and the mixture of their economic activities with natural resources, is fundamental to the environmental justice. To avoid a situation in which only these populations will get negative outcomes from the exploitation of the resources, efforts to commercialize medicinal plants need to be inclusive and ensure they get their fair share of the benefits and their rights upheld. Ethical conduct in research fosters enlightenment and education which ensure the subject is heard and well-informed. It is important that the powers that be acknowledge the importance of sustainable living, respect for indigenous knowledge systems, and more equitable income distribution. The stakeholders have the capacity to develop moral values that raise the standards of the managements of natural resources and give the governments the responsibility of guarding the rights of the pedestals by engaging in socialization and communication with each other.

6.4 Sustainability

Sustainability is where an ethical factor should definitely be considered, when one looks for phototherapies. It is a holistic approach that can include ecological, social, and economic sustainability. Besides, conservation of a wide range of species and promotion of health and prosperity locally and environmentally is the goal of sustainable practices as well as a protection of the future of population of medicinal plants. The ecological balance and the reduction of plant population would be impossible if ecological sustainability won't be maintained for the plants to serve medicinal purposes. An excessive amount of collection of the various sorts of herbal medicine plants indeed goes on in the wild; however unsustainable harvesting methods may eventually lead to the overexploitation of habitats and destruction of habitats. Therefore, researchers should not just set strategic goals but should also put strategies into play. The demonstration of sustainable methods of gathering is one of the strategies to ensure ecologically sustainable development. It means to incorporate the collection of medical plants which enables mother nature to rebound and

keeps the plants to play their ecological roles within the biospheres where they grow. By practicing selective harvesting, that is, focusing on the mature plants while allowing the young guns to thrive, or rotational harvesting, which is a form of harvesting that gives the already harvested regions a chance to regenerate over time, are just two practical sustainable harvesting strategies. However, conservation efforts sustained toward medicinal plant species endangered and their natural environments should be undertaken a well. For the purpose of reducing the pressure on wild populations, conservation techniques such as setting aside protected areas, habitat restoration projects and raised in botanical gardens or in living heath units can be used. A significant number of researchers who are in favor of biodiversity preservation and ecological resistance actively encourage the protection of medicinal plants. One of the crucial factors in preserving social sustainability is to make sure that everyone achieves this benefit fairly, and this applies to the local populations in the area where medicinal plants are discovered since most of their traditional livelihoods ride heavily on using these plants. These communities have unique and potentially sustainable value, where they may be able to achieve economic development, increased salaries, and medical services accessibility. Although the economic bit is of great significance when it comes to the marketing of medicinal plants the provisioning of economic sustainability is another essential component towards the detection of these plants. Sustainable economic activities have to be socially responsible, should be commercially feasible, and also, they should be environment friendly. Such could be putting into practice fair trade policies, as well as the support of neighborhood companies and cooperatives, and the funding for marketing and value-added processing endeavors. The closing stage is working on the green factors that encompass sustainability. By pursuant the ecological, social, and economic sustainability, scientists may forestall the loss of biodiversity, building community well-being, and in the long run, provision for these medicinal plant resource. Stakeholders can safeguard benefits of plant systematization are being fulfilled while as well maintaining the ecosystem's integrity together with future generations’, through the application of sustainable methods.

CHAPTER 7
SUMMARY, CONCLUSION, RECOMMENDATION AND IMPLICATION FOR FUTURE RESEARCH

7.1 Summary of the Study

The research on "Medicinal Plants Detection with Transfer Learning: A Deep CNN-Based Comparison of Detection and Segmentation Approaches" represents a comprehensive exploration of advanced artificial intelligence techniques in the domain of health and agriculture. Leveraging a dataset of 1160 original photos, the study investigates the effectiveness of deep Convolutional Neural Networks (CNNs) in recognizing and segmenting six types of medicinal plant images: “Arjun Leaf”, “Curry Leaf”, “Marsh Pennywort Leaf”, “Mint Leaf”, “Neem Leaf”, “Rubble Leaf”. The research rigorously tests three CNN-based models, achieving notable accuracy scores. Transfer learning is examined, revealing nuances in accuracy when the input picture deviates from the training data. The study underlines the potential of ensemble models over individual architectures and recognizes the impact of dataset size on prediction capabilities, aligning with recommendations for augmentation-based updates. While the research promises advancements in medicinal plants detection, it also acknowledges environmental considerations and articulates a sustainability plan to mitigate associated impacts. The limitation of this study is that it ignores the background problem of the leaf image. Further research will be carried out using a dataset with a complex background by adding a segmentation method before being processed by the powerful model.

7.2 Conclusions

In summary, the research on the identification of medicinal plants utilizing VGG-16, VGG-19, and Custom CNN models has demonstrated notable progress in the intersection of everyday life and technology. Using a dataset of 1160 pictures and an accuracy rate of 97% achieved by the Custom CNN model, the work shows the effectiveness of deep learning models in automating the identification of medicinal plants in other plants. This accomplishment not only represents a significant improvement in the precision and

effectiveness of medicinal plant identification, but it also shows how technology has the power to fundamentally alter human lifestyles. Even beyond the technological advances, ethical considerations have been essential in ensuring responsible technology usage and addressing any societal ramifications. According to the larger discussion on the ethical use of AI in social life, the study highlights the necessity to establish a balance between innovation and ethical responsibility. Through the development of a dependable and effective method for detecting medicinal plants, which will promote the use of medicinal plants for a healthy life and decry the use of chemicals, the ultimate goal of this research is to advance scientific understanding while also having a positive, observable effect on social life in the rapidly evolving field of technological innovation.

7.3 Implication for Future Study

The number of people who are trying to use the computer vision and machine learning in plant science is on the rise, especially in the detection and the monitoring of the medicinal plants. To somehow be able to discover and classify the medicinal plant species, "A Transfer Learning-Based Approach to Detect Medicinal Plants" provides a new and novel approach that makes a big contribution to this field. The research demonstrates how transfer learning could be applied to the feature extraction and utilization on plant classification tasks by using the deep learning models that have previously been trained and then refining them on the dataset that has been put together for the medicinal plants. The transfer learning helps in cutting down the training time and the computational expense and also adds the model's capacity to generalize the plant species and the environmental circumstances. Besides, the research stresses the fact that using the trustworthy metrics of evaluation, such as precision, recall, and F1-score, is so important when the classification model is found to be effective. Through the process of careful testing and check after check, the research proves the reliability and efficiency of the proposed method in the identification of the diverse plant species that has the healing property and the diseases that are connected to them. In the coming times, researchers may study the possibilities of going for a huge dataset, solve the problems of class imbalance, and finally, add a multi-modal data source to the list in order to boost the accuracy and the capability of the transfer

learning technique. Besides, the fact that to prove the model's performance in different environmental situations and encourage its use for plant health monitoring and management, it must be implemented and validated in the real scenarios like botanical gardens and agricultural fields, should not be neglected. On the whole, the study gives new directions for the design of the intelligent systems for the identification of medicinal plants, which will in turn have a significant impact on the sustainable agriculture, traditional medicine, and biodiversity preservation.

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Appendix

Course Outcomes, Complex Engineering Problems (EP), and Complex Engineering Activities (EA) Addressing

Title:

Deep Learning Approaches to Detect Medicinal Plants

Student ID: [201-15-13650]

CO Description for FYDP

CO	CO Descriptions	PO
	Phase -I	
CO1	Integrate recently gained and previously acquired knowledge to identify a **medicinal plant detection** problem for the Final Year Design Project (FYDP)	**PO1**
CO2	Analyze different aspects of the goals in designing a solution for this FYDP	**PO2**
CO3	Explore diverse problem domains through a literature review, delineate the issues, and establish this goal for the FYDP	**PO4**
CO4	Perform economic evaluation and cost estimation and employ suitable project management procedures throughout the development life cycle of the FYDP	**PO11**
	Phase -II	
CO5	Design and develop technical solutions and system components or processes that meet specified requirements, ensuring compliance with public health and safety standards, as well as considering cultural, socioeconomic, and environmental factors in this FYDP	**PO3**

CO6	Choose and apply appropriate methodologies, resources, and contemporary engineering and IT technologies to address complex engineering processes, encompassing prediction and modeling, while adhering to relevant constraints in this FYDP	**PO5**
CO7	Analyze societal, health, safety, legal, and cultural considerations, along with associated responsibilities, in the context of professional engineering practice and the resolution of this problem, employing logical reasoning guided by contextual understanding.	**PO6**
CO8	Comprehend and evaluate the enduring sustainability and impact of professional engineering endeavors in addressing intricate engineering challenges within social and environmental frameworks.	**PO7**
CO9	Implement ethical principles and adhere to professional standards and norms in this FYDP	**PO8**
CO10	Capable of operating proficiently both individually and as a team member or leader across diverse teams and interdisciplinary settings in this FYDP.	**PO9**
CO11	Proficiently communicate with the engineering community and broader society regarding complex engineering endeavors, including the ability to comprehend and generate comprehensive reports and design documentation, as well as provide and receive clear instructions throughout this FYDP.	**PO10**
CO12	Acknowledge the importance of self-directed and life-long learning within the evolving landscape of technology, and possess the readiness and capability to engage in lifelong learning endeavors.	**PO12**

A. Dataset Description: The Dataset consists of consumer data that was collected as a result of conducting a survey. The study with a dataset composed of high-resolution photos of several species of medicinal plants, for example, Arjun, Curry, Marsh Pennywort, Mint, Neem, and Rubble leaves, is based on these photos. The correct types of plants that have been listed on each photograph may result in several examples of leaves with different orientations, backgrounds, and lights. To make sure the model training and assessment are reliable, the dataset is spitted into training, validation and test sets in the appropriate ratios.

B. Transfer Learning Model Architecture: The issue of the Architecture of the Transfer Learning Model is the architectural design of the model that is mainly used to transfer learning from one domain to another domain. This research is based on the transfer learning model architecture that consists of a deep convolutional neural network (CNN) with the architecture that has already been trained like VGG-16, ResNet, or InceptionV3. The weights from a major-scale picture classification challenge, like ImageNet, are employed to initialize the pre-trained model, which is later improved by the medicinal plant dataset. The bottom layer of the network is altered by means of methods like feature extraction or fine-tuning to fit the particular requirements of the plant species classification.

C. Experimental Setup: High-performance GPUs which have been installed on the workstation made it possible for the model training and the assessment to be done efficiently during the trials. Through the preprocessing process, the dataset was altered by the actions of image resizing, pixel value normalization, and training data augmentation in order to make the model general. The hyperparameters such as learning rate, batch size, and optimizer were adapted to the model and the performance was improved by using grid search or random search techniques.

D. Evaluation Metrics: The common classification measures, which included the accuracy, F1-score, precision, and recall of the transfer learning-based method, were used to assess its performance. Remember the formula of recall is the percentage of true positive predictions among all real positive instances, and the one of precision is the percentage of true positive forecasts over all the positive predictions. A good evaluation of the model's performance is provided by the F1-score, which is the harmonic mean of precision and recall. The correctness of this model in all classes is evaluated by accuracy.

E. Results and Discussion: The trial results demonstrated that the transfer learning-based strategy was able to identify and classify the medicinal plant species with great accuracy. The model showed a high level of performance in plant species recognition and disease detection, which is the F1-score, precision, and recall of several classes of plants were excellent. The outcomes' significance was analyzed thereof in the context of the possible

research directions and the possible usage in the field of plant health management and monitoring.

F. Code Implementation: I used Google Colab for writing code. I am learning lot of things to work this research. The extra items present the code for the transfer learning-based strategy, which is the data preprocessing, model training, assessment, and inference. I learn some well-known deep learning frameworks sus as TensorFlow or PyTorch are used in order to implement the code, which also has highly-detailed documentation and guides that can be used for replication and further experimentation.